

SUPPLY CHAIN ANALYTICS REPORT

AtliQ Mart

AI-Powered Supply Chain Analytics & OTIF Performance Monitoring

Order fulfilment, delivery reliability and customer service performance across 35 customers in Ahmedabad, Vadodara and New Jersey

Prepared by

Panicker Nimesh Mahendran

Data Analyst | Portfolio project report

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1. Executive Summary

Business situation

AtliQ Mart is a growing food manufacturer and distributor (dairy, food and beverages) selling to 35 retail customer accounts in Ahmedabad, Vadodara and New Jersey (US). Management's concern is customer service: orders that arrive late, short, or both. This report converts the project's order data into a validated view of delivery reliability and product availability.

Analytical objective

Measure fulfilment at the right grain (order lines versus orders), compare each customer with its own service target, find where failures concentrate, and separate what the data proves from what still needs investigation.

Scope note – read first

The KPI sheet in Output.xlsx (Sheet3) summarises a **1,000-line sample (the first four days of orders)**, and it calculates On-Time, In-Full and OTIF on **order lines**. The uploaded CSV files hold the full period: **24,195 order lines and 13,467 orders** (placed 1 Mar–16 May 2025). The headline figures below use the full data at **order level**, the grain used in the brief's KPI definitions. Workbook values are shown alongside and reconciled in Sections 7 and 18.

Key findings

- **Only 28.7% of orders are delivered On Time In Full** (3,866 of 13,467). The workbook's 45.9% OTIF is a line-level, sample-based figure and overstates order-level performance.
- **Line Fill Rate is 65.9%, but Volume Fill Rate is 96.6%.** Most shortfalls are shallow: shortfall lines averaged 90% of the ordered quantity, and only 3.4% of ordered units went undelivered.
- **On-Time is 59.2% and In-Full is 52.6% of orders.** Against the customer targets supplied in dim_targets_orders.csv (order-weighted: 85.3% / 76.3% / 65.1%), the gap is -26.0, -23.8 and -36.4 points. **None of the 35 customers meets its OT, IF or OTIF target.**
- **Failures concentrate in 12 customers:** 34% of accounts but 40.2% of orders and 33.9% of order value, with a combined OTIF of 13.8% versus 38.7% for the other 23. They split into two different problems: **late delivery** (8 accounts, On-Time about 29%) and **short delivery** (7 accounts, In-Full 16–26%); three accounts have both.
- **Order value is almost evenly spread** (largest customer is only 1.16× the smallest; top 5 hold 15.1% of value), so service risk is a breadth problem rather than a dependence on a few big accounts. Three of the ten highest-value accounts are in the failing group.
- **Performance is flat and uniform across product, city, weekday and week**, so the data points to customer-specific causes rather than a product shortage or a regional issue.



Figure 1 – Actual order-level performance versus customer targets

Management takeaway

- Treat the **28.7% order-level OTIF** as the true baseline; retire the 45.9% line-level sample figure from reporting.
- Start with the **12 failing accounts**: lifting them to the level of the other 23 would raise network OTIF from 28.7% to 38.7% (+10.0 points; scenario, not a forecast).
- Assign different owners: **logistics/dispatch** for the 8 late-delivery accounts, **inventory and allocation** for the 7 short-delivery accounts.
- Even the 23 'core' accounts reach only 38.7% OTIF against an average target near 70%, so a system-wide fix is also needed, not only account firefighting.
- Review whether a small tolerance on shortfalls is commercially acceptable: 62% of short lines are within 10% of the ordered quantity. A scenario in which these count as in-full lifts order OTIF to 45.2%.
- Fix reporting before automating alerts: the workbook mixes grains, has swapped day/month dates, truncated lookups and missing USD valuations (Section 6).

2. Business Problem

AtliQ Mart serves retail chains with short agreed delivery windows: the agreed delivery date is 1, 2 or 3 days after order placement for every order line in the data. In this setting a delivery that is a day late or a few percent short breaks the service promise, and every failure is a customer-service event even when the commercial loss is small.

The supply-chain challenges named in the project brief are order fulfilment, delivery reliability, product availability, customer service and regional or customer-level performance. The data shows these challenges are real but not evenly distributed: they sit in specific customer accounts, as Sections 9–13 show.

3. Project Objective

- Transform operational order data into KPIs that management can act on: Line Fill Rate, Volume Fill Rate, On-Time %, In-Full % and OTIF %.
- Compare actual service with the customer-specific targets provided in the dataset.
- Identify high-value customers with weak service and distinguish late-delivery from short-delivery problems.
- Demonstrate an automated, AI-assisted analytics workflow in which every number is validated, not simply generated.

4. Technical Architecture

The project workflow, as described in the project brief, moves sales data from email or CSV into an automated pipeline and finally into spreadsheet analysis. The report focuses on business analysis; the architecture is summarised for context. The n8n workflows and database configuration were not part of the uploaded files, so they are described, not independently inspected.

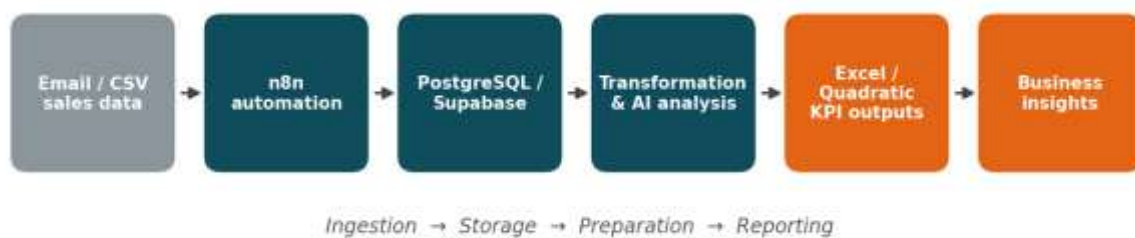


Figure 2 – End-to-end analytics workflow

Layer	Role in the project
Email / CSV sales data	Source files: customer, product, target, order-line and order-level tables (the five CSVs supplied).
n8n automation	Ingests incoming files and orchestrates the load, so refreshes do not depend on manual copying.
PostgreSQL / Supabase	Central structured store for dimension and fact tables; keys (customer_id, product_id, order_id) support joins.
Transformation & AI analysis	Joins, KPI calculation and natural-language-assisted analysis. Output.xlsx is the exported result.
Excel / Quadratic	Spreadsheet outputs: order summary (Sheet2), KPI summary (Sheet3) and top-customer tables (Sheet4, Sheet5).
Business insights	Interpretation, risk segmentation and recommendations (this report).

5. Dataset Overview

Workbook and source files

Source	Content	Size
Output.xlsx – Sheet1	dim_date: date, year, month, day, day name, month name, week of year, weekend flag	92 days (1 Mar–31 May 2025)
Output.xlsx – Sheet2	Order_Summary_INR: order line joined to customer, city, product, category, flags, total_amount	999 lines × 16 columns
Output.xlsx – Sheet3	KPI summary: 7 KPIs (static values, no formulas)	7 rows
Output.xlsx – Sheet4	Top5_Customers: value, OTIF %, IF %, OT % (sample-based)	5 rows
Output.xlsx – Sheet5	Top 5 customers by order value with order counts (full-data based)	5 rows
dim_customers.csv	Customer ID, name, city, currency (INR or USD)	35 customers
dim_products.csv	Product ID, name, category, INR and USD price	18 products
dim_targets_orders.csv	Per-customer OT, IF and OTIF targets (%)	35 rows
fact_order_line.csv	One row per order line: quantities, dates, IF/OT/OTIF flags	24,195 lines
fact_aggregate.csv	One row per order: order-level on_time, in_full, otif flags	13,467 orders

Data model

Customer → Orders → Order Lines → Products, with each order line also carrying placement, agreed and actual delivery dates (linking to dim_date) and each customer linked to its service targets. Delivery performance is measured on order lines and rolled up to orders: an order is on time, in full or OTIF only if **all** its lines are (confirmed: the fact_aggregate flags equal the minimum of the line flags for 100% of the 13,467 orders).

Dimension	Full dataset (CSV files)
Order lines / orders	24,195 lines / 13,467 orders (1.80 lines per order; 56% of orders have a single line)
Customers	35 accounts: 12 in Ahmedabad, 12 in Vadodara (INR billed) and 11 in New Jersey, US (USD billed)
Products	18 SKUs in 3 categories: Dairy (12: milk, butter, ghee, curd), Food (3: biscuits), Beverages (3: tea)
Date range	Placed 1 Mar–16 May 2025; agreed delivery 2 Mar–19 May; actual delivery 1 Mar–22 May 2025
Order quantity per line	20–500 units (median 206); total 5,676,448 units ordered
Order value basis	₹604.87M at order quantity × INR list price (reproduces Sheet5 exactly); US customers are billed in USD
Targets	OT, IF and OTIF targets for all 35 customers (dim_targets_orders.csv); not present in Output.xlsx itself

6. Data Quality Assessment

Two layers were checked: the full CSV files (the data behind the KPIs) and the Output.xlsx deliverable. The CSV layer is clean. The workbook has several issues that affect how its numbers can be read; none changes the full-data results in this report, but each should be fixed before the workbook feeds a dashboard or automated alerts.

Issues found in Output.xlsx

Data Quality Check	Result	Business Impact
Row reconciliation (Sheet2 vs Sheet3)	Issue. Sheet3 reports 1,000 order lines; Sheet2 contains 999. The 1,000th line (order FMR35702301) is missing.	Detail and summary do not reconcile; Sheet3 cannot be reproduced from Sheet2 alone.
Date validity (Sheet2)	Issue. Day and month are swapped: all 999 placement dates fall on the 3rd of Jan–Apr 2025 (source value 01-03-2025 is 1 March, shown as 3 January). The source CSV is valid DD-MM-YYYY.	Any trend, weekday or dim_date join built on Sheet2 is wrong. dim_date covers 1 Mar–31 May only.
Customer lookup (Sheet2)	Issue. 33 lines have blank customer name and city; all belong to customer 789903 (Elite Mart, Vadodara), the last row of dim_customers.	Elite Mart Vadodara disappears from customer or city pivots, a likely lookup range one row short.
Product lookup (Sheet2)	Issue. 52 lines have blank product name and category; all are product 25891603 (AM Tea 100), the last row of dim_products.	Beverages appear as 112 lines instead of 164 in category analysis built on Sheet2.
Missing values: total_amount	Issue. 252 of 999 lines (25.2%) have no amount: 212 of 312 US lines (68%) and 40 INR lines (all AM Tea 100).	Order value is understated, and unevenly: USD-billed customers lose most. Sheet4 rankings are biased as a result.
Sheet4 vs Sheet5 consistency	Issue. Both are labelled top customers with OTIF %, IF % and OT %, but Sheet4 is line-level on the 1,000-line sample and Sheet5 is order-level on the full data. No customer appears in both lists.	Similarly named columns carry different meanings; a reader can quote the wrong figure.
Sheet5 coverage	Issue. Sheet5 lists only INR-billed customers. On the same value basis Foodtown (₹18.25M, rank 2) and Whole Foods Market (₹18.18M, rank 4) would belong in the top 5.	The top-5 list is not the top 5 across all customers, and the scope is not stated.
KPI grain (Sheet3)	Issue. On Time, In Full and OTIF % are line-level. In Full Delivery % (62.3%) equals Line Fill Rate (62.3%): the same measure shown twice.	Order-level equivalents on the full data are 59.2%, 52.6% and 28.7%; the workbook overstates OTIF.
KPI scope (Sheet3)	Issue. The KPIs describe the first 1,000 of 24,195 lines (4.1%), covering 1–4 March 2025.	Not representative of the full period; the label does not say it is a sample.
Calculation anomaly (Sheet3)	Issue. Total Orders is stored as 571.9999999999999; values are hard-coded constants, not formulas.	Floating-point artefact: harmless to the value but shows the calculation path is not auditable in the file.
Category labels	Minor issue. "beverages" is lower-case while "Dairy" and "Food" are capitalised.	Can split a category in case-sensitive tools.

Checks passed (full CSV files)

Data Quality Check	Result	Business Impact
Missing values	Passed. No blanks in any of the 11 columns of fact_order_line or 6 of fact_aggregate.	KPIs are not distorted by missing data.
Duplicates	Passed. No duplicate lines, no repeated (order, product) pair, no repeated order_id in fact_aggregate.	No double counting.
Invalid dates	Passed. All 72,585 date values parse as valid DD-MM-YYYY; no actual or agreed delivery earlier than placement.	Date logic is usable.
Invalid quantities	Passed. No order quantity ≤ 0, no negative delivery, no over-delivery (delivered never exceeds ordered).	Fill rates are bounded and meaningful.
Missing customer / product IDs	Passed. All 35 customer IDs and 18 product IDs exist in the dimension tables; all 35 customers have targets.	Joins are complete.
Flag consistency	Passed. For 100% of 24,195 lines: In Full = (delivered ≥ ordered), On Time = (actual ≤ agreed), OTIF = In Full × On Time.	The business definitions are applied consistently.
Order vs line reconciliation	Passed. fact_aggregate flags match the line flags rolled up (all lines must pass) for all 13,467 orders.	Order-level KPIs are internally consistent.

Checks that could not be performed

- **Inventory and stock-outs:** no stock-level data, so shortfalls cannot be tied to availability.
- **Reason codes:** no cause codes for late or short deliveries.
- **Currency conversion:** no FX table; the rate implied in Sheet2 (about 87.3 INR per USD, 85.3 on 15 lines) cannot be verified.
- **Sample selection logic:** the workbook does not state how its 1,000 lines were chosen (they match the first 1,000 lines of fact_order_line).
- **Target definitions:** the period and calculation basis of dim_targets_orders.csv are not documented.

7. Supply Chain KPI Scorecard

KPI	Actual (full data)	Workbook Sheet3 (sample)	Interpretation	Business Assessment
Total Orders	13,467	572	Distinct order IDs, 1 Mar–16 May 2025	Full volume is 23× the sample
Total Order Lines	24,195	1,000	Order-line records	Sample is 4.1% of lines
Line Fill Rate (lines)	65.9%	62.3%	Lines delivered in full ÷ total lines	About 1 line in 3 is short
Volume Fill Rate (units)	96.6%	96.5%	Units delivered ÷ units ordered	Shortfalls are shallow: 3.4% of units
On-Time Delivery % (orders)	59.2%	71.8% (lines)	Orders with every line delivered by the agreed date	Below the 85.3% order-weighted target
In-Full Delivery % (orders)	52.6%	62.3% (lines)	Orders with every line fully delivered	Below the 76.3% target
OTIF % (orders)	28.7%	45.9% (lines)	Orders on time and in full	Below the 65.1% target; about 1 in 3.5 orders succeeds

Target benchmark: No target benchmark exists in Output.xlsx. The targets used here come from dim_targets_orders.csv, which holds a target for each customer. The order-weighted averages are OT 85.3%, IF 76.3% and OTIF 65.1%. The OTIF target is close to OT × IF (65.1%), which suggests it was derived from the other two. The workbook's percentages are stored as decimals (for example 0.623) and are shown here as percentages.

8. KPI Interpretation

Line Fill Rate

Definition: fully fulfilled order lines ÷ total order lines = $15,952 \div 24,195 = 65.9\%$. **Why it matters:** it measures SKU-level availability for each customer request, independent of how many lines an order contains. **Reading:** about one line in three ships short. This is the line-level counterpart of In-Full and is the same number as the workbook's 'In Full Delivery %'.

Volume Fill Rate

Definition: delivered quantity ÷ ordered quantity = $5,483,166 \div 5,676,448 = 96.6\%$. **Why it matters:** it shows whether demand is met by volume. **Reading:** the company ships almost all the volume requested. The gap between a 96.6% volume fill and a 65.9% line fill means failures are frequent but shallow; they are driven by the strict in-full rule (any shortfall fails the line) more than by large supply gaps.

On-Time Delivery %

Definition: orders delivered on or before the agreed date ÷ total orders = $7,975 \div 13,467 = 59.2\%$. At line level the figure is 71.2%. **Why it matters:** it measures delivery reliability against the customer's promise.

Reading: 4 in 10 orders have at least one late line. Early deliveries (1 day early, 6.3% of lines) count as on time.

In-Full Delivery %

Definition: orders in which every line is delivered at the ordered quantity ÷ total orders = $7,078 \div 13,467 = 52.6\%$. **Why it matters:** it measures whether the customer received the whole order. **Reading:** only about half of orders arrive complete. It is lower than the line fill rate because one short line fails the whole order.

OTIF %

Definition: orders both on time and in full ÷ total orders = $3,866 \div 13,467 = 28.7\%$. **Why it matters:** OTIF is among the strongest customer-service indicators because it combines timing and quantity. **Reading:** fewer than three orders in ten meet the full service promise. At line level the same data gives 47.8%, the grain behind the workbook's 45.9%.

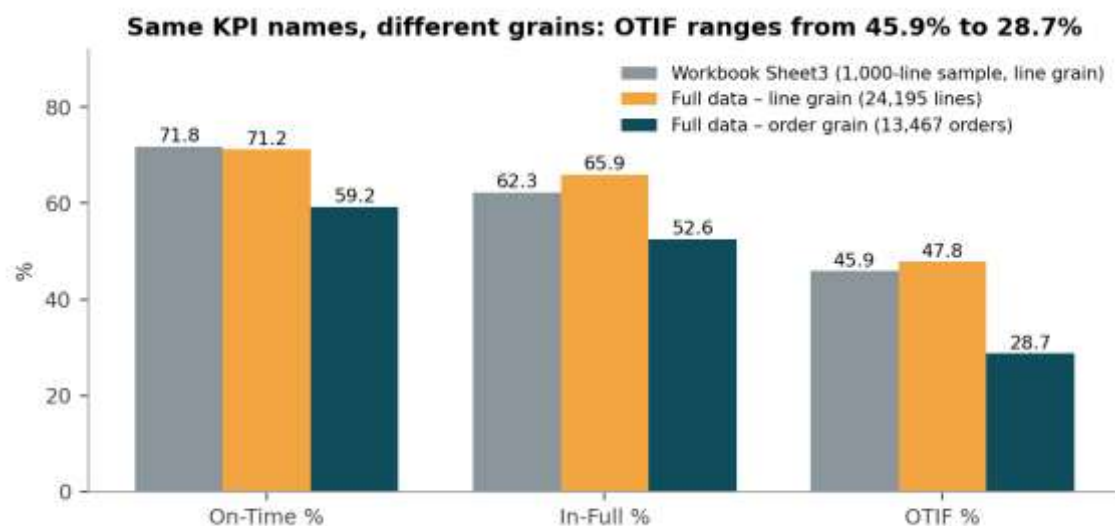


Figure 3 – The same data produces different KPIs depending on the grain

Metric	Sheet3 (sample, line)	Sample, order level*	Full data, line	Full data, order
On-Time %	71.8%	61.5%	71.2%	59.2%
In-Full %	62.3%	47.6%	65.9%	52.6%
OTIF %	45.9%	27.7%	47.8%	28.7%

*Recalculated from the 999 lines present in Sheet2 (571 orders).

9. Customer Performance Analysis

Workbook view: top 5 customers by order value (Sheet5, full data)

Rank	Customer	Customer ID	City	Order Value (INR)	OTIF %	IF %	OT %	Orders
1	Propel Mart	789402	Ahmedabad	18,534,490	39.1%	56.4%	74.5%	353
2	Elite Mart	789902	Ahmedabad	18,194,928	40.1%	58.2%	74.8%	349
3	Vijay Stores	789102	Ahmedabad	18,154,402	35.6%	58.8%	69.5%	371
4	Acclaimed Stores	789521	Ahmedabad	18,117,153	20.1%	66.8%	30.1%	518
5	Vijay Stores	789103	Vadodara	17,970,915	9.1%	15.8%	75.1%	342

All figures are order-level and match an independent recalculation from the CSV files. As noted in Section 6, Sheet5 contains only INR-billed customers. Note that the fifth customer, Vijay Stores Vadodara, has an In-Full rate of 15.8%.

Top 10 customers by order value, all 35 accounts (same value basis)

Rank	Customer	ID	City	Order Value (INR)	OTIF %	IF %	OT %	OTIF target	Segment
1	Propel Mart	789402	Ahmedabad	18,534,490	39.1%	56.4%	74.5%	71%	Core
2	Foodtown	789301	New Jersey, US	18,247,182	30.5%	55.8%	68.7%	69%	Core
3	Elite Mart	789902	Ahmedabad	18,194,928	40.1%	58.2%	74.8%	73%	Core
4	Whole Foods Market	789320	New Jersey, US	18,184,193	36.7%	58.5%	70.3%	74%	Core
5	Vijay Stores	789102	Ahmedabad	18,154,402	35.6%	58.8%	69.5%	73%	Core
6	Acclaimed Stores	789521	Ahmedabad	18,117,153	20.1%	66.8%	30.1%	62%	On-Time failure
7	Vijay Stores	789103	Vadodara	17,970,915	9.1%	15.8%	75.1%	62%	In-Full failure
8	Viveks Stores	789503	Vadodara	17,950,486	38.9%	62.7%	69.0%	71%	Core
9	Info Stores	789603	Vadodara	17,922,188	43.7%	64.9%	71.6%	69%	Core
10	Lidl	789420	New Jersey, US	17,770,867	21.7%	67.9%	28.8%	59%	On-Time failure

Value is order quantity × INR list price, the basis that reproduces Sheet5. On this basis Foodtown and Whole Foods Market (both US) rank 2nd and 4th. If US orders were valued at USD prices converted at the rate implied in Sheet2 (about 87.3 INR per USD), US accounts would rank higher still; their ranks here are conservative.

Concentration and spread

- Order value ranges from ₹15.94M to ₹18.53M per customer; the largest is only **1.16×** the smallest and the coefficient of variation is 3.9%.
- The top 5 customers hold **15.1%** of value and the top 10 hold **29.9%**: low concentration.
- OTIF varies far more than value: from **5.9%** (Fairway Market, New Jersey) to **48.5%** (Propel Mart, Vadodara), an 8× spread.
- Every customer is below its OTIF target. The shortfall ranges from 19.5 points (Propel Mart, Vadodara) to 53.2 points (Price Rite, New Jersey).

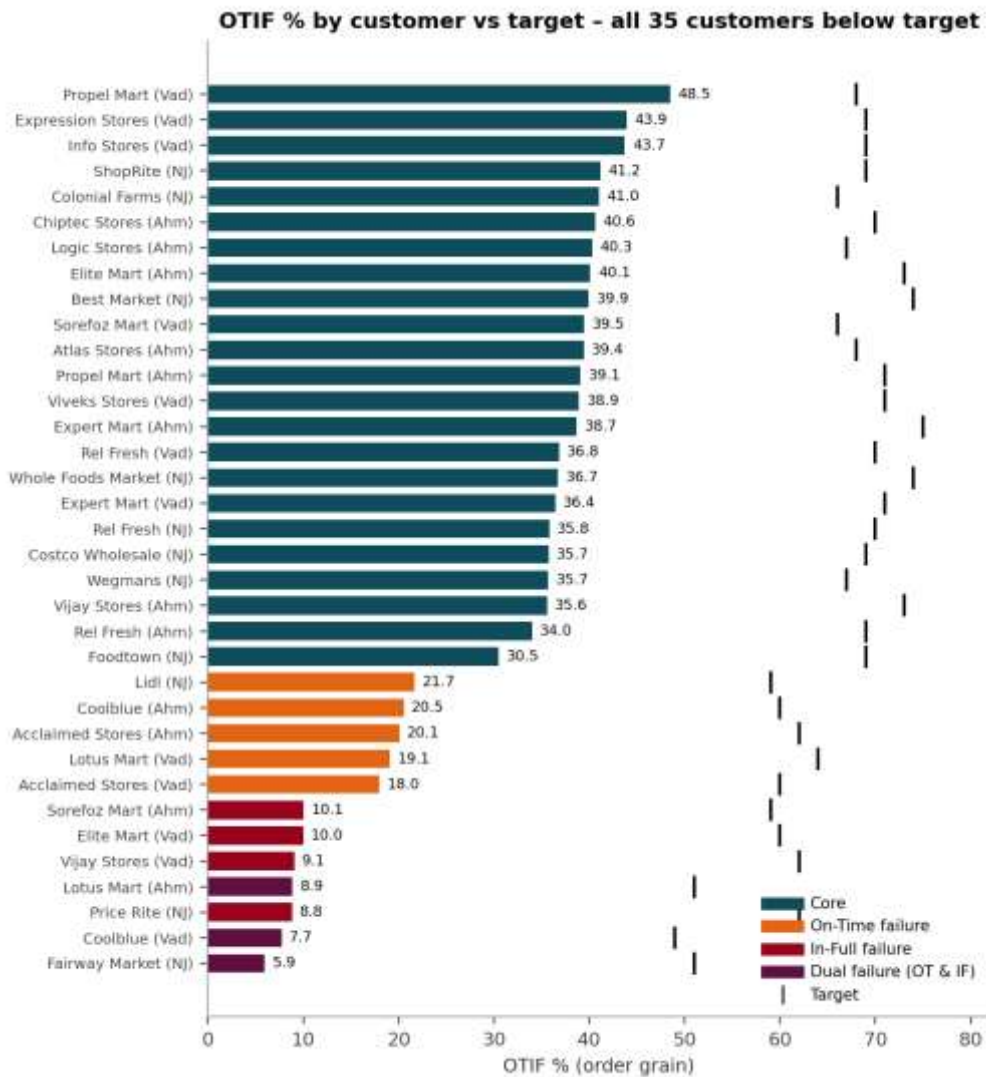


Figure 4 – OTIF % by customer versus customer target (order grain)

Workbook view: Sheet4 top 5 (1,000-line sample)

Rank	Customer	ID	City	Order Value (INR)	OTIF %	IF %	OT %
1	Best Market	789501	New Jersey, US	1,098,117	67.7%	71.0%	93.5%
2	Logic Stores	789721	Ahmedabad	939,466	60.0%	63.3%	86.7%
3	Rel Fresh	789203	Vadodara	929,734	72.4%	75.9%	89.7%
4	Info Stores	789603	Vadodara	900,258	69.2%	84.6%	80.8%
5	Colonial Farms	789720	New Jersey, US	891,978	64.3%	75.0%	85.7%

Sheet4 is based on roughly 30 lines per customer, with percentages calculated on lines. Best Market's value of 1,098,117 is built from only 11 of its 31 lines because USD amounts are missing; the ranking is therefore unreliable and the table is shown for completeness only. This report uses the full-data figures.

10. High-Value Customer Risk Analysis

Value is nearly flat across customers, so 'high value' here means above the median (₹17.29M) or in the top 10. The more useful question is which commercially important accounts are failing, and how badly relative to their own targets.

Value rank	Customer	Order value	OT %	IF %	OTIF %	Target	Gap (pts)	Failure type
6	Acclaimed Stores (Ahm)	₹18.12M	30.1%	66.8%	20.1%	62%	-41.9	On-Time failure
7	Vijay Stores (Vad)	₹17.97M	75.1%	15.8%	9.1%	62%	-52.9	In-Full failure
10	Lidl (NJ)	₹17.77M	28.8%	67.9%	21.7%	59%	-37.3	On-Time failure
12	Price Rite (NJ)	₹17.60M	71.3%	17.4%	8.8%	62%	-53.2	In-Full failure
18	Elite Mart (Vad)	₹17.29M	73.4%	16.9%	10.0%	60%	-50.0	In-Full failure

The table lists the failing accounts in the top half of the value ranking. The remaining seven failing accounts rank 24th–34th by value and are listed in Appendix A.

- **Acclaimed Stores, Ahmedabad (rank 6): high value + weak On-Time.** On-Time is 30.1% against a 78% target; In-Full is normal (66.8%). Orders arrive late rather than short, which points to a delivery-scheduling problem. Its sister account in Vadodara shows the same pattern.
- **Vijay Stores, Vadodara (rank 7): high value + weak In-Full.** In-Full is 15.8% against a 67% target, while On-Time is 75.1% (target 92%). Its Ahmedabad sister account is far healthier (OTIF 35.6%), so the cause appears to be local to the Vadodara account.
- **Lidl, New Jersey (rank 10): high value + weak On-Time** (28.8% against a 75% target). **Price Rite, New Jersey (rank 12): high value + weak In-Full** (17.4% against a 67% target; OTIF 8.8%).
- **Why this matters:** each of these accounts buys about ₹16–18M over the period, the same as the best-served accounts, yet receives a fraction of the service. A commercially important account with weak OTIF is exposed to service-level deterioration, penalties, delisting or lost shelf space; the data cannot say whether any of this has occurred.

Customer segmentation

Method. Each customer is classified on its own order-level On-Time % and In-Full %. Customers fall into two tight clusters on each measure with an empty gap between them: On-Time is either 28–32% or 69–78% (nothing between 31.5% and 68.7%), and In-Full is either 16–26% or 54–69% (nothing between 26.0% and 53.6%). The thresholds **On-Time < 50%** and **In-Full < 40%** sit in these gaps, so the classification does not depend on the exact cut-off. Value is not used as a threshold because it varies by only 16% across customers; a value split is shown separately below.

Segment	Cust.	Orders	Order value	OT	IF	OTIF	Avg target	Lines/order	Avg order value
Core	23	8,048 (60%)	₹399.77M (66%)	72%	60%	38.7%	70%	1.98	₹49.7K
On-Time failure	5	2,523 (19%)	₹85.66M (14%)	29%	67%	19.9%	61%	1.36	₹34.0K
In-Full failure	4	1,411 (10%)	₹69.76M (12%)	73%	17%	9.5%	61%	1.97	₹49.4K
Dual failure	3	1,485 (11%)	₹49.70M (8%)	29%	23%	7.5%	50%	1.37	₹33.5K

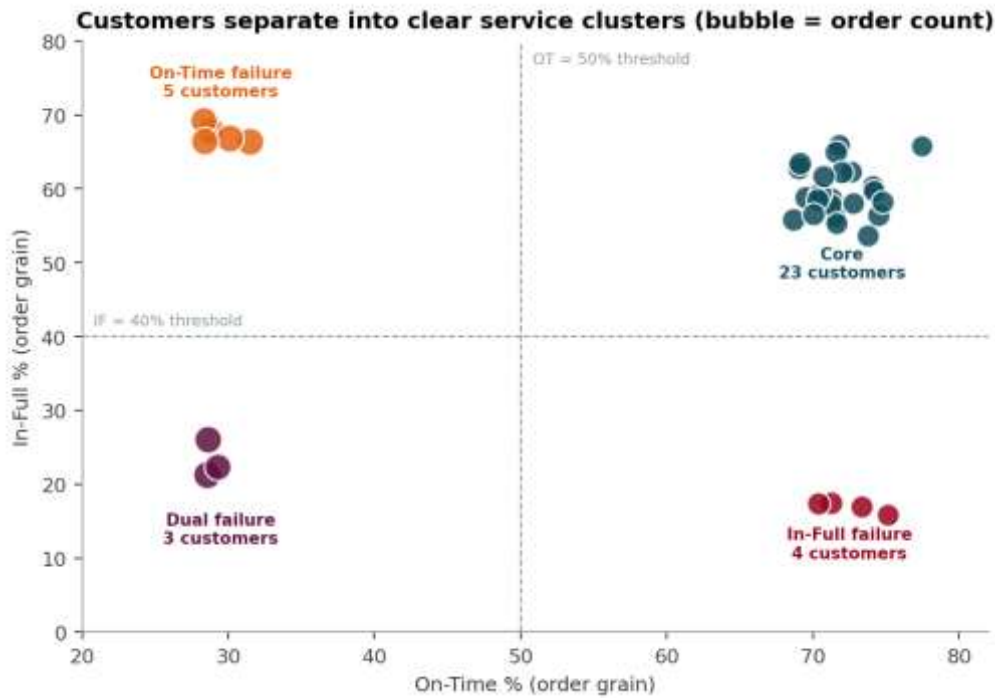


Figure 5 – Customer clusters on On-Time % and In-Full % (order grain)

Value × service view. Splitting at median value: high value with relatively strong service 13 customers; **high value with a service problem 5** (Acclaimed Ahmedabad, Vijay Vadodara, Lidl, Price Rite, Elite Mart Vadodara); lower value with relatively strong service 10; **lower value with a service problem 7**. 'Strong service' means the Core segment, which still misses every target (Section 13). With such small value differences this 2×2 is indicative only.

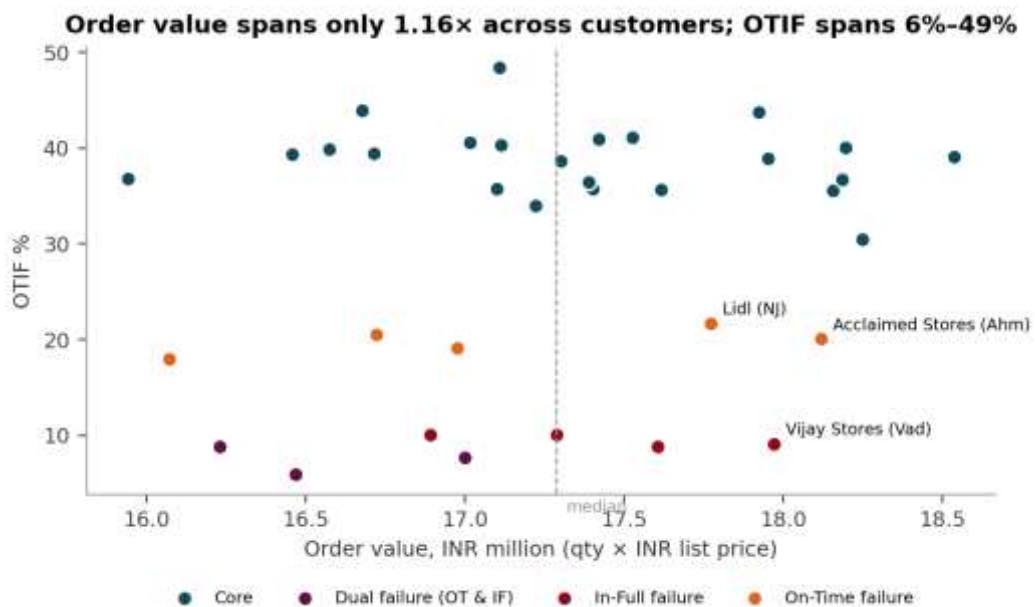


Figure 6 – Order value versus OTIF by customer: no relationship between commercial size and service

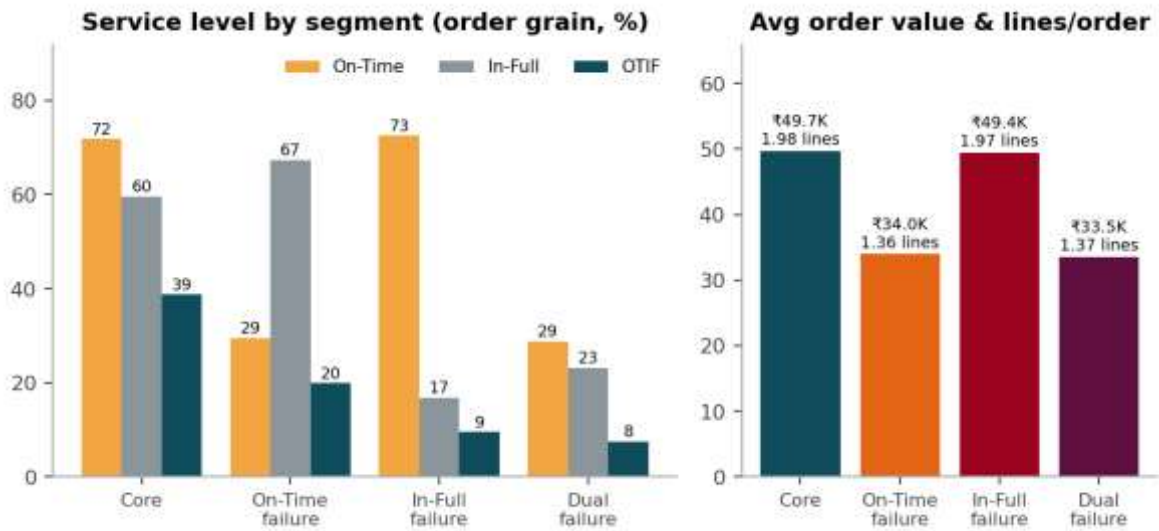


Figure 7 – Service level, order size and basket structure by segment

11. Geographic Analysis

Geography	Cust.	Orders	Lines	Value (INR list)	Line fill	Volume fill	OT %	IF %	OTIF %
India	24	9,338	16,564	₹412.96M	65.8%	96.6%	58.4%	52.9%	28.5%
USA	11	4,129	7,631	₹191.91M	66.3%	96.6%	61.2%	51.8%	29.1%
– Ahmedabad	12	4,732	8,388	₹207.96M	67.4%	96.8%	58.5%	54.1%	29.3%
– Vadodara	12	4,606	8,176	₹205.01M	64.0%	96.4%	58.2%	51.7%	27.7%
– New Jersey, US	11	4,129	7,631	₹191.91M	66.3%	96.6%	61.2%	51.8%	29.1%

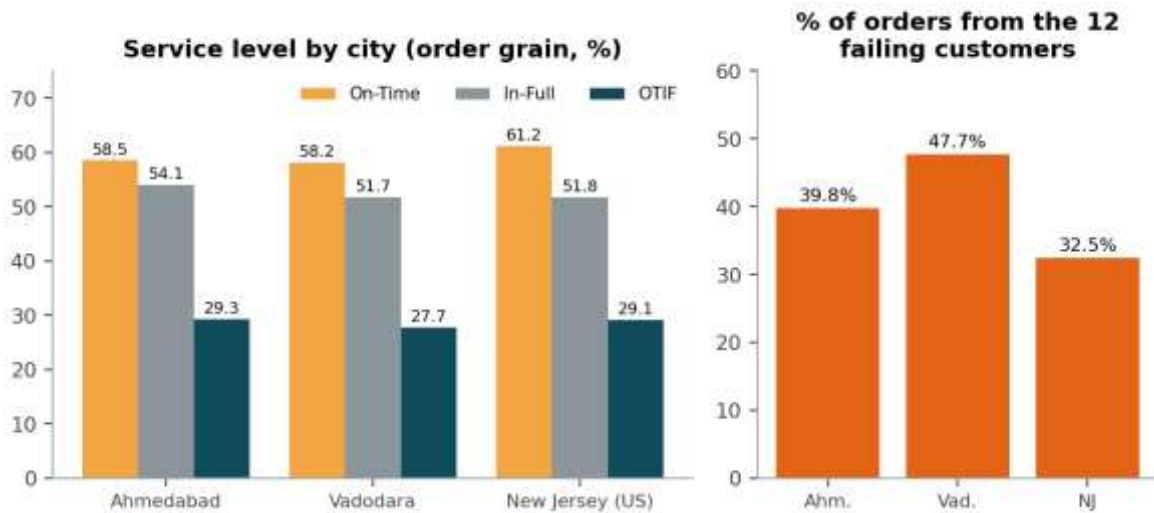


Figure 8 – Service by city, and share of each city's orders coming from the 12 failing customers

- **India versus USA is essentially a tie:** OTIF 28.5% versus 29.1%. The USA is better on-time (61.2% vs 58.4%); India is slightly better in-full (52.9% vs 51.8%). The differences are small and no country-level problem is evident.

- **Vadodara is the weakest city** (OTIF 27.7%), with the lowest line fill rate (64.0%). It hosts 5 of the 12 failing accounts, which supply 48% of its orders, versus 32% in New Jersey. The city gap reflects customer mix more than a city-wide cause.
- **Commercial importance:** Ahmedabad and Vadodara each carry about 34% of list-price value and New Jersey about 32% on the INR list basis; valued at USD prices the US share would be considerably higher (Section 9).
- The data holds no delivery-route, warehouse or carrier fields, so geographic service risk cannot be traced below city level.

12. Product / Category Analysis

Category	Lines	Qty ordered	Qty delivered	Value (INR list)	Line fill	Volume fill	OT % (line)	OTIF % (line)
Dairy	16,086	4,453,599	4,301,946	₹396.71M (66%)	65.8%	96.6%	70.9%	47.6%
Food	4,095	720,325	695,746	₹143.52M (24%)	66.2%	96.6%	72.0%	48.6%
Beverages	4,014	502,524	485,474	₹64.65M (11%)	66.2%	96.6%	71.8%	48.1%

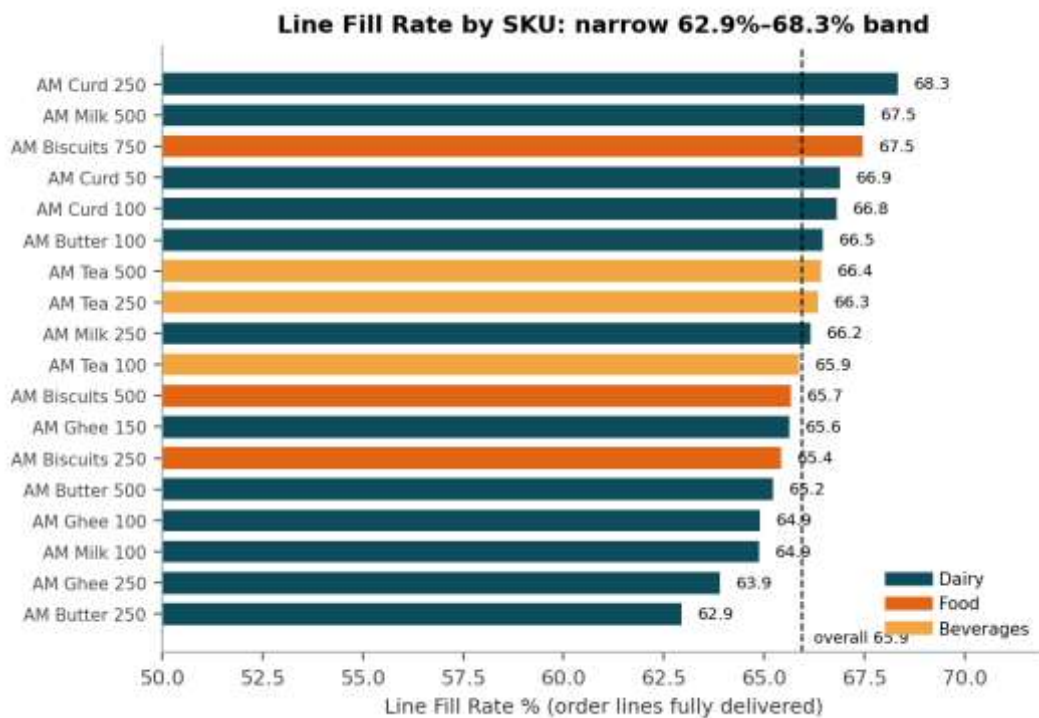


Figure 9 – Line Fill Rate by SKU: a narrow band around the 65.9% average

- **No product or category stands out.** Line Fill Rate is 65.8–66.2% in all three categories and 62.9%–68.3% across the 18 SKUs. With about 1,300 lines per SKU, random variation alone is roughly ± 2.5 points, so the differences between SKUs are consistent with noise.
- **Dairy dominates the portfolio:** 66% of value and 78% of units ordered, so any dairy availability issue would matter most. The lowest line fill rates are AM Butter 250 (62.9%) and AM Ghee 250 (63.9%), the highest AM Curd 250 (68.3%).
- Because the shortfall pattern is the same in every SKU, the data does not support a product-specific supply constraint. Stock-level data would be needed to test one.

13. Root-Cause Analysis

The data is observational: it records outcomes, not causes. The table separates what the data shows from what remains a hypothesis.

Observed facts

#	Observation	Evidence
1	Failure is account-specific	12 of 35 customers have OTIF of 5.9–21.7% while the other 23 range 30.5–48.5%. Customers sit in two tight clusters on both OT and IF (Section 10).
2	Not driven by product	Line fill rate 62.9–68.3% across all 18 SKUs; 65.8–66.2% across categories.
3	Not driven by city, weekday or time	OTIF by city 27.7–29.3%; by weekday 27.9%–29.7%; by week 27.0%–29.8% over 10 complete weeks. No improvement or deterioration.
4	Not driven by agreed lead time	Line-level OTIF is 47.0%, 48.0% and 48.5% for 1-, 2- and 3-day agreed lead times.
5	Late deliveries are systematic, not rare	In the 8 late-delivery accounts 73% of lines are late, spread over +1 day (30%), +2 days (27%) and +3 days (17%). No line is ever more than 3 days late.
6	Late-delivery accounts order differently	They average 1.37 lines and ₹33.8K per order versus 1.98 lines and ₹49.7K for core accounts, and place about 501 orders each versus 350.
7	Shortfalls follow three depth tiers	Short lines deliver about 95% (40%), 90% (39%) or 80% (21%) of the ordered quantity. The depth distribution is the same in every segment; what differs is how often a line is short: line fill is about 30% in the short-delivery accounts versus 75% in core.
8	Targets were already lower for the problem accounts	The 8 late-delivery accounts have OT targets averaging 77% (core: 88%) and the four short-delivery-only accounts have IF targets of 65–67% (core: 79%). Actual results are still about 46–51 points below even these relaxed targets.

Potential hypotheses (not confirmed)

- **Delivery windows or cut-offs:** a potential contributor to the late-delivery accounts may be account-specific delivery windows or order cut-off times that do not match dispatch schedules; this should be investigated further.
- **Small, frequent drops:** the late-delivery accounts place more orders with fewer lines each; the data indicates a link between this order pattern and lateness but cannot show whether consolidation would help.
- **Allocation rules:** the three fixed shortfall tiers (about 5%, 10%, 20%) look more like rule-based allocation or rounding than random stock-outs, and the short-delivery accounts are short on 70% of lines. This should be investigated further, for example whether stock is allocated by customer priority or in fixed pack or batch sizes.
- **Account-level data or process issues:** because the same account shows the problem across all products and all weeks, order-entry, master-data or customer-specific handling should be checked.
- The available dataset does not allow us to confirm any of these: it has no stock levels, dispatch times, routes, carriers or reason codes.

The Core accounts are not healthy either

The 23 Core customers average 38.7% OTIF against an average target of 70%; their line fill rate is 75.1% and their on-time rate is 71.8%. The same shortfall tiers appear in them. This points to a system-wide availability and delivery limit in addition to the 12 account-specific problems.

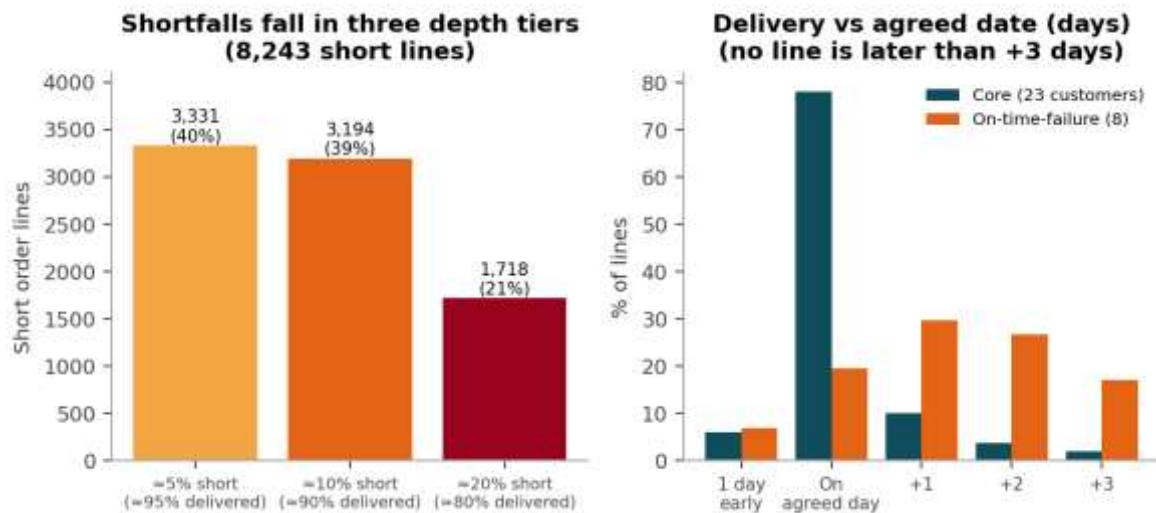


Figure 10 – Shortfall depth tiers and timing versus agreed date

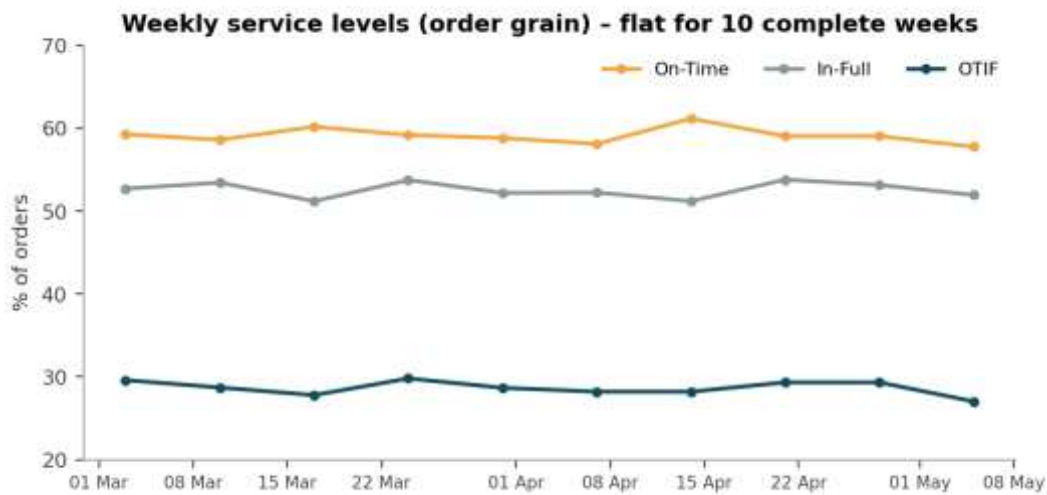


Figure 11 – Weekly order-level service: no trend

14. Key Business Insights

Insights are ordered by business importance.

#	Finding	Evidence	Business Impact	Recommended Action
1	Real order-level OTIF is 28.7%, not 45.9%	3,866 of 13,467 orders are OTIF; workbook 45.9% is line-level on a 4% sample. 72% of list-price value (₹433.41M) sits in orders that failed OTIF.	Management believes service is better than it is; improvement targets set on the wrong base.	Re-baseline reporting on order-level, full-period OTIF; label every KPI with its grain.
2	No customer meets any target	0 of 35 customers meet OT, IF or OTIF targets. Network gap: OT -26.0, IF -23.8, OTIF -36.4 points.	The service problem is system-wide; targets are not met even at the best accounts.	Pair account fixes with a network-level availability and delivery plan; review whether targets are realistic.
3	12 accounts drive the failure cluster	34% of accounts, 40% of orders, 34% of value; OTIF 13.8% vs 38.7% elsewhere. Lifting them to Core level raises network OTIF 28.7% → 38.7%.	Largest single improvement available, and it is concentrated in a manageable list.	Run a 12-account recovery programme with named owners and weekly OTIF reviews.

#	Finding	Evidence	Business Impact	Recommended Action
4	Two different failure modes need two different owners	5 accounts fail on-time only (OT ≈ 29%, IF normal); 4 fail in-full only (IF ≈ 17%, OT normal); 3 fail both (OT 29%, IF 23%).	One generic fix will not work; the cause sits in dispatch for some accounts and in stock allocation for others.	Logistics owns the 8 late-delivery accounts; inventory/order management owns the 7 short-delivery accounts.
5	Shortfalls are shallow but the in-full rule is strict	VFR 96.6% vs LFR 65.9%; 62% of short lines are within 10% of the order. Scenario: counting ≤10% shortfall as in-full gives LFR 87.1%, order OTIF 45.2%.	Many failures are cosmetic in volume terms but still break the service promise and the KPI.	Agree with key customers what tolerance, if any, is contractual; focus on closing the 20% tier first.
6	Late deliveries are small but systematic	73% of lines in the 8 accounts are late, all by 1–3 days. Same accounts: 1.37 lines and ₹34K per order, ~501 orders each.	Short, repeated delays suggest scheduling rather than breakdowns, and frequent small drops add dispatch pressure.	Review delivery windows and dispatch sequencing; test consolidated or fixed-day delivery for these accounts.
7	High-value accounts are among the failing ones	Acclaimed Stores Ahmedabad (rank 6, OTIF 20.1%), Vijay Stores Vadodara (rank 7, 9.1%), Lidl (rank 10, 21.7%), Price Rite (rank 12, 8.8%).	Large accounts get small-account service; relationship and contract risk.	Put these four on an executive-sponsored watch list with customer-specific OTIF targets.
8	Value is evenly spread, so risk is broad	Largest customer 1.16× smallest; top 5 = 15.1%. No relationship between value and OTIF.	There is no small group of accounts to protect; every account is equally important.	Manage service by exception across all 35 accounts rather than by size tier.
9	Performance is flat and uniform across dimensions	OTIF by week 27.0%–29.8%; by city 27.7–29.3%; SKU LFR 62.9–68.3%.	No self-correction, and no product or region to blame; the cause sits in customer-level processes.	Do not wait for improvement; investigate customer-specific order, dispatch and allocation practice.
10	The reporting layer needs fixing before automation	Swapped dates, one-row-short lookups (customer 789903, product 25891603), 25% blank order values, mixed grains in Sheet3, Sheet4 and Sheet5.	Automated dashboards and alerts would publish wrong or inconsistent numbers.	Correct the issues in Section 6, then build alerts on the validated order-level definitions.

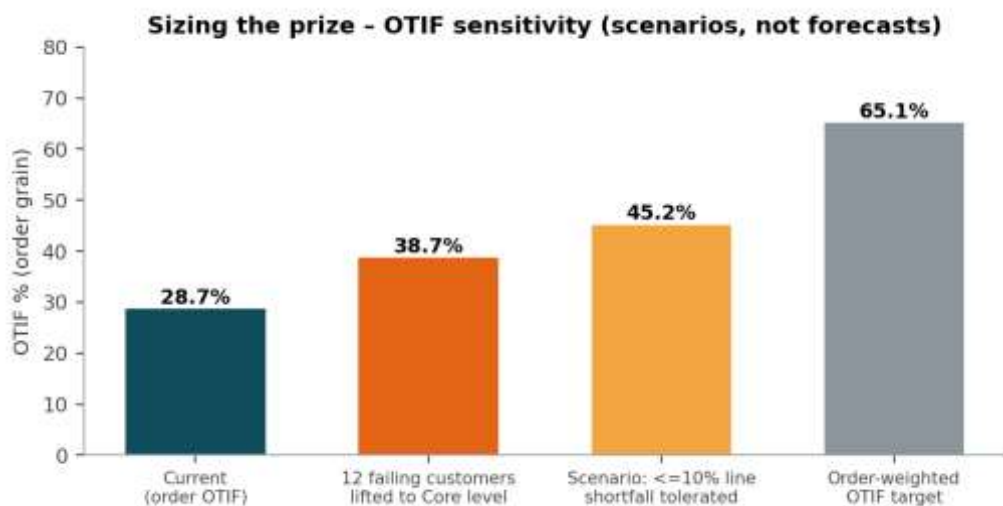


Figure 12 – OTIF scenarios. Illustrative sensitivities calculated from the full dataset; not forecasts

15. Recommendations

Area	Recommendation	Linked finding
Inventory management	Investigate why short lines cluster at 5%, 10% and 20% shortfall and whether allocation is rule-based. Analyse the short-delivery accounts (Vijay Vadodara, Price Rite, Elite Vadodara, Sorefoz Ahmedabad, Coolblue Vadodara, Fairway, Lotus Ahmedabad) first. Introduce safety-stock and allocation policy only after stock-level data is added.	Insights 4, 5
Order fulfilment	Prioritise allocation for orders from the failing accounts and high-value orders. Close the 20% shortfall tier first. Agree a documented tolerance for minor shortfalls with key customers.	Insights 3, 5, 7
Logistics	Review delivery windows, dispatch cut-offs and sequencing for the 8 late-delivery accounts (Acclaimed Ahmedabad and Vadodara, Lidl, Lotus Vadodara and Ahmedabad, Coolblue Ahmedabad and Vadodara, Fairway). Pilot fixed-day or consolidated delivery for accounts placing many small orders.	Insights 4, 6
Customer management	Set customer-specific OTIF targets and a monthly service review for the 12 failing accounts; put the four high-value accounts (Acclaimed Ahmedabad, Vijay Vadodara, Lidl, Price Rite) under executive sponsorship.	Insights 3, 7, 8
Analytics & automation	Correct the workbook issues; publish order-level KPIs with grain labels; automate weekly KPI reports, a customer-level OTIF alert against the target, and exception lists for any account whose OT or IF falls below 50% or 40%.	Insights 1, 2, 10

16. Dashboard Recommendations

KPI cards: Total Orders, Total Order Lines, Line Fill Rate, Volume Fill Rate, On-Time %, In-Full %, OTIF %, each shown with its grain (line or order) and with its target where available.

Chart	Business question it answers	Design note
1. OTIF by customer	Which accounts are furthest from target?	Sorted bars with target marker; colour by segment (Figure 4).
2. Order value by customer	Where is the commercial weight?	Bar chart; expect a flat profile (Section 9). Show currency basis.
3. OTIF by city / region	Is there a geographic pattern?	Include the share of orders from failing accounts (Figure 8).
4. On-Time vs In-Full comparison	Is the problem late, short, or both?	Scatter by customer with thresholds (Figure 5); the key diagnostic.
5. Fill-rate performance	Are shortfalls frequent or deep?	Line fill vs volume fill, plus shortfall-tier bars (Figure 10).
6. Top customers by order value	How are the largest accounts served?	Top 10 table with OTIF and gap to target.
7. High-value customers with low OTIF	Which important accounts are at risk?	Value-vs-OTIF scatter with watch-list labels (Figure 6).

Add a weekly OTIF trend with the target line, and a slicer for segment and city. Keep Line Fill Rate and OTIF visually distinct so line-level and order-level measures are never confused.

17. AI & Automation Benefits

Component	Benefit
n8n	Automates file ingestion and workflow orchestration, removing manual refresh steps and reducing version drift.
PostgreSQL / Supabase	Provides one structured source with enforced keys; makes joins and re-aggregation reproducible.
AI-powered spreadsheet / analysis tools	Natural-language-assisted cleaning, calculation and charting. Speeds up first-pass analysis considerably.
Python / SQL	Used to validate and recalculate KPIs reproducibly. In this report, Python (pandas) was used to recompute every KPI from the raw CSV files.

AI accelerates analysis, but this project also shows its limits: the workbook produced by the automated steps contains mixed grains, swapped dates and truncated lookups that only domain knowledge and independent checking caught. Speed gains are real only when the validation steps in Section 18 are part of the workflow.

18. Technical Validation

Validation step	What was done	Result
Formula checks	Recomputed LFR, VFR, OT, IF, OTIF from fact_order_line and fact_aggregate.	LFR 65.9%, VFR 96.6%, order OTIF 28.7%; reproduces fact_aggregate exactly.
Joins	Joined facts to customer, product and target dimensions; checked unmatched keys.	No unmatched IDs in the CSVs; Sheet2 lookups drop customer 789903 and product 25891603.
Aggregation level	Compared line-level and order-level KPIs on the same data.	OTIF 47.8% (line) vs 28.7% (order); a 19-point difference.
Duplicate counting	Tested duplicate lines, repeated order–product pairs and repeated orders.	None found. Order-level counts use distinct order IDs.
Percentages	Checked storage format and rebuilt the figures from counts.	Workbook stores decimals (0.623); recalculated values agree with Sheet3 on the sample (62.3%, 71.8%, 45.9%).
Missing values	Counted blanks in every column in the CSVs and the workbook.	CSVs complete; Sheet2 has blank names, categories and amounts (Section 6).
Definitions	Re-derived IF, OT and OTIF flags from quantities and dates.	100% agreement on 24,195 lines; early delivery counts as on time.
Cross-check of reported tables	Rebuilt Sheet5 from raw data (value, orders, OTIF, IF, OT).	All five rows match exactly; Sheet4 matches only the sample, at line level.

Why the grain matters

Supply-chain metrics are easy to misstate when the denominator changes. An order with four lines is one order but four lines. At line level, a customer whose orders each contain one short line out of four still shows 75% fill; at order level the same orders are all failures. In this dataset 44% of orders have two or more lines, so the two grains diverge strongly: OTIF is 47.8% by lines and 28.7% by orders. The right grain is the one the customer experiences, which is the order. Each KPI should be labelled with its denominator, and line-level and order-level figures should never share a column name.

19. Limitations

- **Short period:** about 11 weeks (1 March–16 May 2025); seasonality and long-run trends cannot be assessed.
- **No inventory data:** shortfalls cannot be linked to stock-outs, production or replenishment.
- **No logistics variables:** no route, carrier, dispatch time, warehouse or reason code; late delivery causes cannot be tested.
- **Targets outside the workbook:** targets exist only in dim_targets_orders.csv, with no documented definition or period.
- **Valuation basis:** value is quantity × INR list price; USD-billed customers are billed in USD and an FX rate cannot be verified, so cross-currency value comparisons are approximate.
- **Observational data:** the analysis can show where failures concentrate, not prove why. Scenarios in Figure 12 are sensitivities, not forecasts.
- **Workbook sample:** Output.xlsx KPIs describe a 1,000-line sample with known defects, so the full CSV files were used for the headline analysis.

20. Future Improvements

Priority	Improvement	Reason (limitation addressed)
1	Add inventory and stock-out tracking at SKU and customer level	No stock data; needed to test the allocation hypothesis for short deliveries.
2	Add reason codes for late and short deliveries	Root causes cannot currently be confirmed.
3	Automated OTIF alerts and exception-based reporting at order grain	Failures concentrate in a few accounts and do not self-correct.
4	Real-time supply-chain dashboard on validated definitions	Replaces the manual, mixed-grain workbook.
5	Delivery-route and warehouse performance analytics	Late-delivery accounts need dispatch-level data.
6	Customer churn-risk signals using OTIF trends and order frequency	High-value accounts with weak service are exposed.
7	Demand forecasting and predictive inventory planning	Becomes valuable once stock and a longer history exist.
8	Supplier performance tracking	Upstream availability is currently invisible.

21. Conclusion

AtliQ Mart ships almost all the volume customers order (96.6%), but only 28.7% of orders reach the customer both on time and in full, and no customer meets its targets. The problem is neither random nor evenly spread: 12 accounts, representing 40% of orders, show failures of two different kinds, while product, city, weekday and week make little difference. The immediate priorities are to re-baseline reporting at order level, assign owners to the late-delivery and short-delivery accounts, and investigate why shortfalls fall into fixed depth tiers. A tolerance review and fixing the workbook's data issues will make future automated monitoring trustworthy.

22. Portfolio Project Summary

Project title

AI-Powered Supply Chain Analytics & OTIF Performance Monitoring

Business problem

AtliQ Mart, a fictional food manufacturer, struggles with order fulfilment and delivery reliability across 35 retail customers in India and the US. Management lacked a validated, customer-level view of On-Time, In-Full and OTIF performance against targets.

Solution

An automated pipeline moves order data from CSV/email through n8n into PostgreSQL/Supabase, where it is transformed and analysed with AI-assisted tools into Excel/Quadratic outputs. KPIs were then independently recalculated from 24,195 order lines at both line and order level, reconciled to the workbook and segmented by customer.

Tools & technologies

- n8n, PostgreSQL, Supabase, Excel, Quadratic and AI-assisted analytics tools (as described in the project brief)
- Python (pandas) for the validation and charts in this report. SQL is listed in the brief 'where applicable'; include it on a portfolio only if it was actually used in the project.

Key KPIs

Line Fill Rate 65.9%; Volume Fill Rate 96.6%; On-Time 59.2%, In-Full 52.6% and OTIF 28.7% (order level, full data).

Key business outcome

Identified that order-level OTIF is 28.7% (not the 45.9% in the workbook), that 12 of 35 customers account for a disproportionate failure cluster with two distinct failure modes, that shortfalls are shallow but follow three fixed tiers, and that the reporting layer needed correction before automation.

Appendix A – Customer Scorecard (all 35 customers)

Rk	Customer	ID	City	Orders	Value ₹M	OT %	IF %	OTIF %	Tgt	Gap	Segment
1	Propel Mart	789402	Ahmedabad	353	18.53	74.5%	56.4%	39.1%	71	-31.9	Core
2	Foodtown	789301	NJ, US	364	18.25	68.7%	55.8%	30.5%	69	-38.5	Core
3	Elite Mart	789902	Ahmedabad	349	18.19	74.8%	58.2%	40.1%	73	-32.9	Core
4	Whole Foods Market	789320	NJ, US	354	18.18	70.3%	58.5%	36.7%	74	-37.3	Core
5	Vijay Stores	789102	Ahmedabad	371	18.15	69.5%	58.8%	35.6%	73	-37.4	Core
6	Acclaimed Stores	789521	Ahmedabad	518	18.12	30.1%	66.8%	20.1%	62	-41.9	On-Time
7	Vijay Stores	789103	Vadodara	342	17.97	75.1%	15.8%	9.1%	62	-52.9	In-Full
8	Viveks Stores	789503	Vadodara	365	17.95	69.0%	62.7%	38.9%	71	-32.1	Core
9	Info Stores	789603	Vadodara	359	17.92	71.6%	64.9%	43.7%	69	-25.3	Core
10	Lidl	789420	NJ, US	507	17.77	28.8%	67.9%	21.7%	59	-37.3	On-Time
11	Wegmans	789401	NJ, US	342	17.62	71.6%	55.3%	35.7%	67	-31.3	Core
12	Price Rite	789601	NJ, US	362	17.60	71.3%	17.4%	8.8%	62	-53.2	In-Full

Rk	Customer	ID	City	Orders	Value ₹M	OT %	IF %	OTIF %	Tgt	Gap	Segment
13	ShopRite	789220	NJ, US	340	17.53	74.1%	60.3%	41.2%	69	-27.8	Core
14	Colonial Farms	789720	NJ, US	366	17.42	69.1%	63.4%	41.0%	66	-25.0	Core
15	Costco Wholesale	789101	NJ, US	361	17.40	71.5%	56.2%	35.7%	69	-33.3	Core
16	Expert Mart	789622	Vadodara	354	17.39	70.1%	56.5%	36.4%	71	-34.6	Core
17	Expert Mart	789621	Ahmedabad	357	17.30	72.0%	62.2%	38.7%	75	-36.3	Core
18	Elite Mart	789903	Vadodara	349	17.29	73.4%	16.9%	10.0%	60	-50.0	In-Full
19	Rel Fresh	789202	Ahmedabad	347	17.22	73.8%	53.6%	34.0%	69	-35.0	Core
20	Logic Stores	789721	Ahmedabad	352	17.11	70.7%	61.6%	40.3%	67	-26.7	Core
21	Propel Mart	789403	Vadodara	324	17.11	77.5%	65.7%	48.5%	68	-19.5	Core
22	Rel Fresh	789201	NJ, US	324	17.10	71.3%	58.6%	35.8%	70	-34.2	Core
23	Chiptec Stores	789321	Ahmedabad	347	17.02	72.6%	62.2%	40.6%	70	-29.4	Core
24	Coolblue	789122	Vadodara	518	17.00	28.6%	21.2%	7.7%	49	-41.3	OT+IF
25	Lotus Mart	789422	Vadodara	487	16.98	28.3%	69.2%	19.1%	64	-44.9	On-Time
26	Sorefoz Mart	789702	Ahmedabad	358	16.89	70.4%	17.3%	10.1%	59	-48.9	In-Full
27	Coolblue	789121	Ahmedabad	511	16.72	31.5%	66.3%	20.5%	60	-39.5	On-Time
28	Sorefoz Mart	789703	Vadodara	337	16.72	74.2%	59.6%	39.5%	66	-26.5	Core
29	Expression Stores	789303	Vadodara	337	16.68	71.8%	65.9%	43.9%	69	-25.1	Core
30	Best Market	789501	NJ, US	338	16.57	72.8%	58.0%	39.9%	74	-34.1	Core
31	Fairway Market	789520	NJ, US	471	16.47	29.3%	22.3%	5.9%	51	-45.1	OT+IF
32	Atlas Stores	789221	Ahmedabad	373	16.46	70.5%	59.2%	39.4%	68	-28.6	Core
33	Lotus Mart	789421	Ahmedabad	496	16.23	28.6%	26.0%	8.9%	51	-42.1	OT+IF
34	Acclaimed Stores	789522	Vadodara	500	16.07	28.4%	66.4%	18.0%	60	-42.0	On-Time
35	Rel Fresh	789203	Vadodara	334	15.94	71.3%	57.8%	36.8%	70	-33.2	Core

Value = order quantity × INR list price. OT, IF and OTIF are order-level. Tgt = OTIF target (dim_targets_orders.csv); Gap = actual – target in percentage points. Segment: Core, On-Time (late delivery), In-Full (short delivery), OT+IF (both).

Appendix B – Definitions and Method

- **Order line** = one product on one order. **Order** = distinct order_id.
- **In Full (line)** = delivered quantity ≥ ordered quantity. **On Time (line)** = actual delivery date ≤ agreed delivery date. **OTIF (line)** = both.
- **Order-level On Time, In Full and OTIF** = the order passes only if all its lines pass (matches fact_aggregate).
- **Line Fill Rate** = fully delivered lines ÷ total lines. **Volume Fill Rate** = $\sum$ delivered quantity ÷ $\sum$ ordered quantity.
- **Order value** = $\sum$ ordered quantity × INR list price (dim_products), matching Sheet5.
- **Targets** are order-weighted averages of customer targets (dim_targets_orders.csv).
- **Segments** use order-level On-Time < 50% and In-Full < 40% (thresholds placed in the empty gaps between customer clusters).
- **Scenarios:** (a) the 12 failing accounts at the Core group's order-level OTIF (38.7%); (b) lines delivering at least 90% of the ordered quantity counted as in-full, then re-rolled to orders. Both are sensitivities, not forecasts.